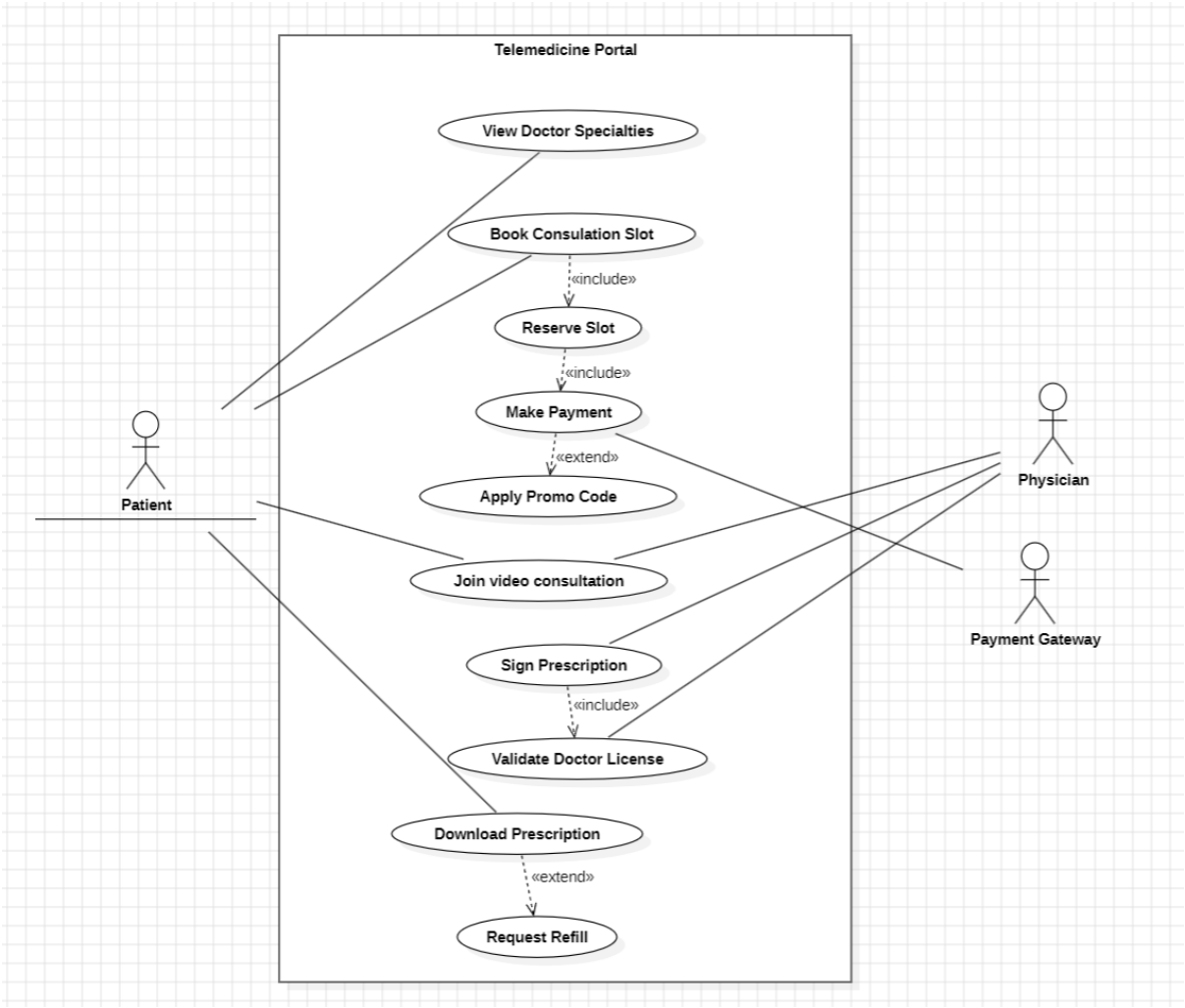
Requirements Table

ReqID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions, automatically compiled into a downloadable PDF.	High	Pass: valid cryptographic signature attached to PDF. Fail: prescription generated without doctor license validation	Core clinical deliverable of the portal.	License validation required.
FR-002	Functional	The system shall allow patients to search and filter doctors by specialty and available time slots.	High	Pass: search results update within 2 seconds and only show doctors matching selected specialty/availability filters.	Entry point of the booking flow — nothing downstream works without doctor discovery.	Supports doctor discovery.
FR-003	Functional	The system shall place a selected slot into a *Reserved* state, locked exclusively to that patient for a fixed timeout window, and auto-release it to *Available* if not confirmed within that window.	High	Pass: a second patient attempting to select the same slot within the timeout is blocked; slot reverts to Available after timeout expires with no payment.	Prevents double-booking; isolates reservation logic from payment logic for independent testing.	Prevents double-booking.
FR-004	Functional	The system shall process the consultation fee payment and transition a Reserved slot to Confirmed only upon successful payment authorization; on failure, the slot reverts to Available per FR-003.	High	Pass: booking status changes to Confirmed only after a payment-gateway success callback. Fail: any Confirmed status without a matching authorization record.	Payment is the gating condition for a real booking; must be independently verifiable from reservation.	Requires payment authorization.
FR-005	Functional	The system shall generate and send an encrypted tele-consultation room link to both patient and physician only after a booking reaches Confirmed status, and no later than a fixed interval before the appointment.	Medium	Pass: room link timestamp is always later than the payment-confirmation timestamp. Fail: link generated for a non-Confirmed booking.	Explicitly named in the problem context; depends on FR-004's confirmation state.	Requires confirmed booking.
NFR-001	Non Functional	All video session metadata and patient clinical notes shall be encrypted in transit (TLS 1.3) and at rest (AES-256).	High	Pass: benchmarking tests confirm target latency and security standards under simulated peak load.	Healthcare data is sensitive by default; non-negotiable baseline.	Applies to healthcare data.
NFR-002	Non Functional	The video consultation service shall establish a connection within 5 seconds on a 4 Mbps network link, verified under simulated load.	High	Pass: 95% of simulated session starts connect within 5 seconds. Fail: median connection time exceeds threshold.	Covers a different axis (speed, not security) than NFR-001 — avoids overlap while still testable.	Applies to video connection.

Use-Case Diagram



Use-Case Flow Specification

Goal	Book Consultation Slot
Actor	Patient with help from Physician and Payment Gateway
Main Flow	1. Patient selects View Doctor Specialties. 2. System displays available doctor specialties. 3. Patient selects a specialty and filters doctors by specialty and available time slots. 4. System displays matching physicians and available consultation slots. 5. Patient selects a physician and an available consultation slot. 6. System places the selected slot in Reserved state and locks it for the patient. 7. System displays the consultation fee and requests payment. 8. Patient optionally applies a Promo Code. 9. System applies the valid promotional discount. 10. Patient selects a payment method. 11. System connects to the Payment Gateway. 12. Payment is successfully authorized. 13. System changes the slot status from Reserved to Confirmed. 14. System confirms the consultation booking. 15. Booking process is completed.
Alternate Flow	Payment Declined: 12a. If payment authorization fails, the system displays a payment failure message and asks the patient to retry. 12b. If the patient successfully retries payment before the reservation timeout, the system proceeds to Step 13. 12c. If payment is not successfully completed before the reservation timeout, the system releases the reserved slot. 12d. The slot status changes from Reserved to Available, and the booking is not confirmed.